

1992 HKCEE MATHS Paper II

1 $\frac{1}{a} + \frac{1}{b} =$

A. $\frac{a+b}{ab}$

B. $\frac{ab}{a+b}$

C. $\frac{1}{ab}$

D. $\frac{2}{a+b}$

E. $\frac{1}{a+b}$

2 If $a = 1 - \frac{1}{1-b}$, then $b =$

A. $1 - \frac{1}{1-a}$

B. $1 - \frac{1}{1+a}$

C. $1 + \frac{1}{1-a}$

D. $1 + \frac{1}{1+a}$

E. $-1 + \frac{1}{1-a}$

3 For what value(s) of x does the equality

$$\frac{(x+1)(x-2)}{x-2} = x+1 \text{ hold?}$$

A. -1 only

B. 2 only

C. Any value

D. Any value except -1

E. Any value except 2

4 $\frac{\sqrt{5}+1}{\sqrt{5}-1} - \frac{\sqrt{5}-1}{\sqrt{5}+1} =$

A. 0

B. $\frac{1}{2}$

C. 3

D. $\sqrt{5}$

E. $\frac{1}{2} + \sqrt{5}$

5 If $\log_{10} b = 1 + \frac{1}{2} \log_{10} a$, then $b =$

A. $10\sqrt{a}$

B. $10 + \sqrt{a}$

C. $5a$

D. $\frac{a}{2}$

E. $1 + \frac{a}{2}$

6 Which of the following is a factor of $4(a+b)^2 - 9(a-b)^2$?

A. $5b - a$

B. $5a + b$

C. $-a - b$

D. $13b - 5a$

E. $13a - 5b$

7 If $\frac{a}{b} = \frac{c}{d} = k$ and a, b, c, d are positive, then which of the following *must* be true?

A. $\frac{a+c}{b+d} = k$

B. $ab = cd = k$

C. $ac = bd = k$

D. $a = c = k$

E. $\frac{ac}{bd} = k$

8

Simplify $\frac{\overbrace{n \times n \times \dots \times n}^{n \text{ times}}}{\underbrace{n + n + \dots + n}_{n \text{ terms}}}.$

A. n^{n-2}

B. $n^{\frac{n}{2}}$

C. $n - 2$

D. $\frac{n}{2}$

E. 1

- 9 If a and b are greater than 1, which of the following statements is/are true?

I. $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

II. $(a^{-1} + b^{-1})^{-1} = a + b$

III. $a^2 b^3 = (ab)^6$

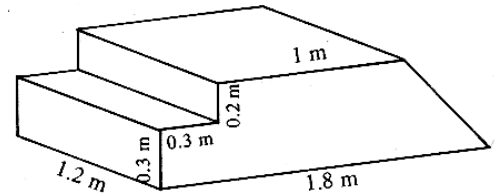
- A. I only
 B. II only
 C. III only
 D. I and II only
 E. None of them
- 10 If $a:b=2:3$, $a:c=3:4$ and $b:d=5:2$, find $c:d$.

- A. 1 : 5
 B. 16 : 45
 C. 10 : 3
 D. 20 : 9
 E. 5 : 1
- 11 Suppose x varies directly as y^2 and inversely as z . Find the percentage increase of x when y is increased by 20% and z is decreased by 20%.

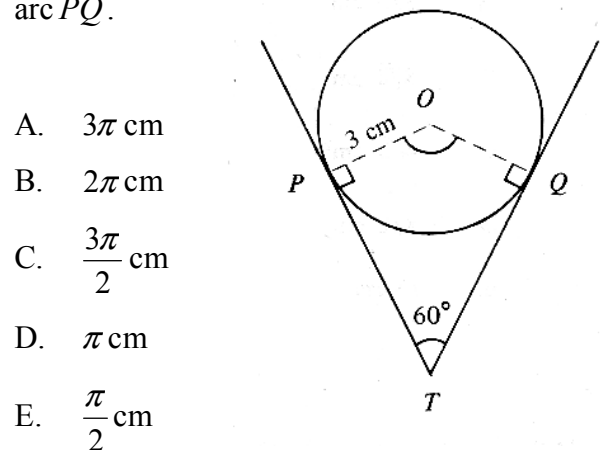
- A. 15.2%
 B. 20%
 C. 50%
 D. 72.8%
 E. 80%
- 12 A sum of \$10000 is deposited at 4% p.a., compounded yearly. Find the interest earned in the *second year*.

- A. \$16
 B. \$400
 D. \$800

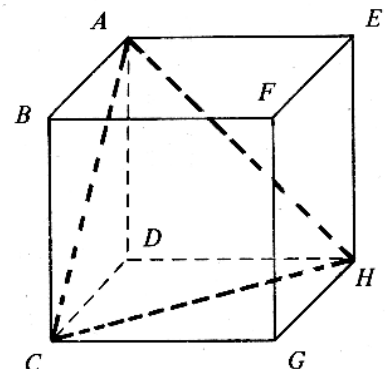
- 13 The figure shows a solid platform with steps on one side and a slope on the other. Find its volume.



- A. 0.75 m^3
 B. 0.84 m^3
 C. 0.858 m^3
 D. 1.008 m^3
 E. 1.608 m^3
- 14 In the figure, TP and TQ are tangent to the circle of radius 3 cm. Find the length of the minor arc PQ .



- A. $3\pi \text{ cm}$
 B. $2\pi \text{ cm}$
 C. $\frac{3\pi}{2} \text{ cm}$
 D. $\pi \text{ cm}$
 E. $\frac{\pi}{2} \text{ cm}$
- 15 Find the ratio of the volume of the tetrahedron $ACHD$ to the volume of the cube $ABCDEFGH$ in the figure.

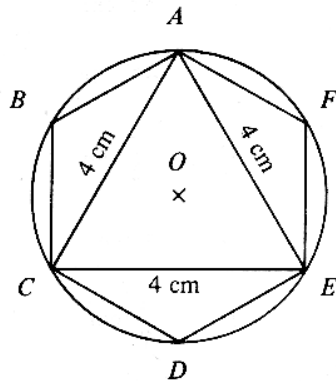


- A. 1 : 8
 B. 1 : 6
 C. 1 : 4

D. 1 : 3

E. 1 : 2

- 16 In the figure, the equilateral triangle ACE of side 4 cm is inscribed in the circle. Find the area of the inscribed regular hexagon $ABCDEF$.

A. $8\sqrt{3}$ cm²B. $8\sqrt{2}$ cm²C. $4\sqrt{3}$ cm²D. $4\sqrt{2}$ cm²E. 16 cm²

- 17 In the figure, a cone of height $3h$ is cut by a plane parallel to its base into a smaller cone of height h and a frustum. Find the ratio of the volume of the smaller cone to the volume of the frustum.

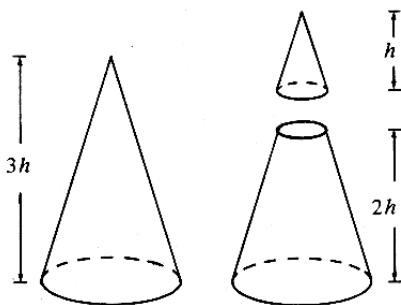
A. 1 : 27

B. 1 : 26

C. 1 : 9

D. 1 : 8

E. 1 : 7



- 18 The greatest value of $1 - 2\sin\theta$ is

A. 5

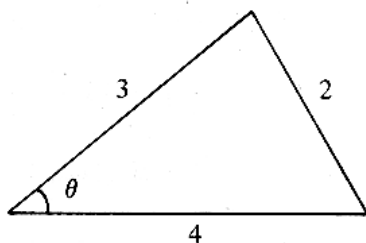
B. 3

D. 0

C. 1

E. -1

- 19 In the figure, find $\cos\theta$.

A. $-\frac{1}{4}$ 

- 20 In which two quadrants will the solution(s) of $\sin\theta\cos\theta < 0$ lie?

A. In quadrants I and II only

B. In quadrants I and III only

C. In quadrants II and III only

D. In quadrants II and IV only

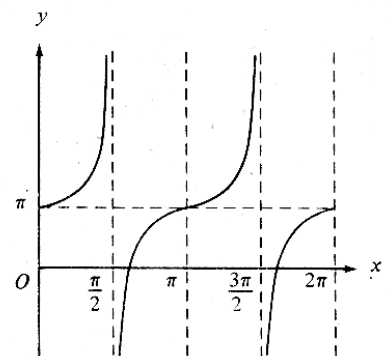
E. In quadrants III and IV only

- 21 If $A + B + C = 180^\circ$, then $1 + \cos A \cos(B + C) =$

A. 0

B. $\sin^2 A$ D. $1 + \sin A \cos A$ C. $1 + \cos^2 A$ E. $1 - \sin A \cos A$

- 22 The figure shows the graph of the function

A. $\tan(x + \pi)$ B. $\tan(x - \pi)$ C. $\pi \tan x$ D. $\pi + \tan x$ E. $\pi - \tan x$ 

- 23 Which of the following equations has/have solutions?

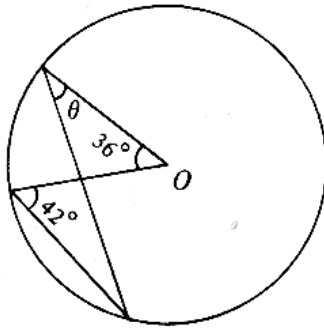
I. $2\cos^2\theta - \sin^2\theta = 1$ II. $2\cos^2\theta - \sin^2\theta = 2$ III. $2\cos^2\theta - \sin^2\theta = 3$

A. I only

- B. II only D. I and II only
C. III only E. II and III only

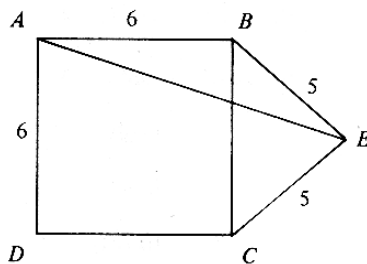
24 In the figure, O is the center of the circle find θ .

- A. 42°
B. 36°
C. 24°
D. 21°
E. 18°



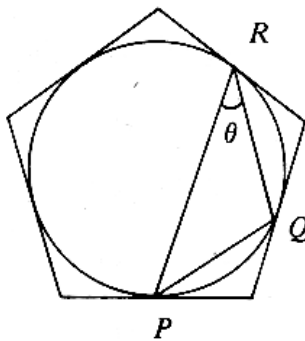
25 In the figure, ABC is a right-angled triangle, $BE=CE=5$, find AE .

- A. $\sqrt{61}$
B. 9
C. 10
D. $6\sqrt{3}$
E. $\sqrt{109}$



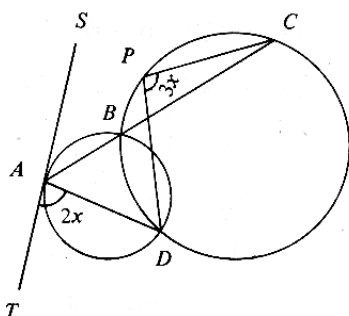
26 In the figure, the circle is inscribed in a regular pentagon. P , Q and R are points of contact. Find θ .

- A. 30°
B. 32°
C. 35°
D. 36°
E. 45°



27 In the figure, ST is a tangent to the smaller circle. ABC is a straight line. If $\angle TAD = 2x$ and $\angle DPC = 3x$, find x .

- A. 30°
B. 36°
C. 40°
D. 42°



28 If the two lines $2x - y + 1 = 0$ and $ax + 3y - 1 = 0$ do not intersect, then $a =$

- A. -6
B. -2 D. 3
C. 2 E. 6

29 If $0 < k < h$, which of the following circles intersect(s) the y-axis?

- I. $(x-h)^2 + (y-k)^2 = k^2$
II. $(x-h)^2 + (y-k)^2 = h^2$
III. $(x-h)^2 + (y-k)^2 = h^2 + k^2$

- A. I only
B. II only D. I and II only
C. III only E. II and III only

30 If the line $y = mx + 3$ divides the circle $x^2 + y^2 - 4x - 2y - 5 = 0$ into two equal parts, find m .

- A. $-\frac{1}{4}$
B. -1
C. 0
D. $\frac{5}{4}$
E. 2

31 The mid-points of the sides of a triangle are $(3,4)$, $(2,0)$ and $(4,2)$. Which of the following

points is a vertex of the triangle?

- A. (3.5,3)
- B. (3,2)
- C. (3,1)
- D. (1.5,2)
- E. (1,2)

- 32 The table shows the mean marks of two classes of students in a mathematics test

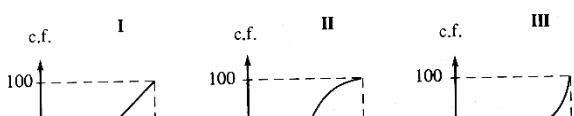
	Number of student	Mean mark
Class <i>A</i>	38	72
Class <i>B</i>	42	54

A student in Class *A* has scored 91 marks. It is found that his score was wrongly recorded as 19 in the calculation of the mean mark for Class *A* in the above table. Find the correct mean mark of the 80 students in the two classes

- A. 61.65
 - B. 62.55
 - C. 63
 - D. 63.45
 - E. 63.9
- 33 Two cards are drawn randomly from five cards *A, B, C, D* and *E*. Find the probability that card *A* is drawn while card *C* is not.

- A. $\frac{3}{25}$
- B. $\frac{3}{20}$
- C. $\frac{4}{25}$
- D. $\frac{6}{25}$
- E. $\frac{3}{10}$

- 34 The figure shows the cumulative frequency curves of three distributions. Arrange the three distributions in the order of their standard



- 35 If the quadratic equation $ax^2 - 2bx + c = 0$ has two equal roots, which of the following is/are true?

- I. a, b, c form an arithmetic sequence.
- II. a, b, c form a geometric sequence.

III. Both roots are $\frac{b}{a}$.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

- 36 Which of the following intervals *must* contain a root of $2x^3 - x^2 - x - 3 = 0$?

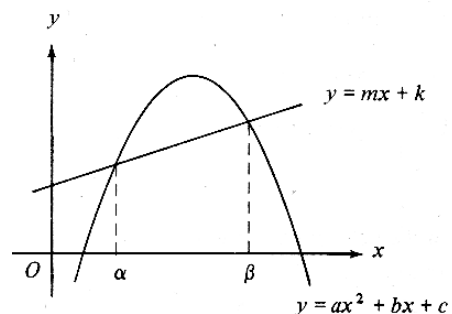
- I. $-1 < x < 1$
- II. $0 < x < 2$
- III. $1 < x < 3$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

- 37 How many integers x satisfy inequality $6x^2 - 7x - 20 \leq 0$?

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

- 38 From the figure, if $\alpha \leq x \leq \beta$, then



- 39 Under which of the following conditions *must* the mean of n consecutive positive integers also be an integer?
- A. $ax^2 + (b-m)x + (c-k) \leq 0$
 B. $ax^2 + (b-m)x + (c-k) < 0$
 C. $ax^2 + (b-m)x + (c-k) = 0$
 D. $ax^2 + (b-m)x + (c-k) > 0$
 E. $ax^2 + (b-m)x + (c-k) \geq 0$
- 40 The L.C.M. of P and Q is $12ab^3c^2$. The L.C.M. of X , Y and Z is $30a^2b^3c$. What is the L.C.M. of P , Q , X , Y and Z ?
- A. $360a^3b^6c^3$
 B. $60a^2b^3c^2$
 C. $60ab^3c^2$
 D. $6a^2b^3c$
 E. $6ab^3c$
- 41 If a polynomial $f(x)$ is divisible by $x-1$, then $f(x-1)$ is divisible by
- A. $x-2$
 B. $x+2$
 C. $x-1$
 D. $x+1$
 E. x
- 42 Find the $(2n)$ th term of the G.S.
- $-\frac{1}{2}, 1, -2, 4, \dots$
- A. 2^{2n}
 B. -2^{2n}
 C. -2^{2n-3}
 D. 2^{2n-2}
 E. -2^{2n-2}
- 43 If the price of an orange rises by \$1, then 5 fewer oranges could be bought for \$100. Which of the following equations gives the original price \$ x of an orange?
- A. $\frac{100}{x+1} = 5$
 B. $\frac{100}{x+1} - \frac{100}{x} = 5$
 C. $\frac{100}{x} - \frac{100}{x+1} = 5$
 D. $\frac{100}{x-1} - \frac{100}{x} = 5$
 E. $\frac{100}{x} - \frac{100}{x-1} = 5$
- 44 By selling an article at 10% discount off the marked price, a shop still makes 20% profit. If the cost price of the article is \$19800, then the marked price is
- A. \$21600
 B. \$26136
 C. \$26400
 D. \$27225
 E. \$27500
- 45 Coffee A and coffee B are mixed in the ratio $x : y$ by weight. A costs \$50/kg and B costs \$40/kg. If the cost of A is increased by 10% while that of B is decreased by 15%, the cost of the mixture per kg remains unchanged.

Find $x : y$.

A. $2 : 3$

B. $5 : 6$

D. $3 : 2$

C. $6 : 5$

E. $55 : 34$

46 In the figure, find $\tan \theta$.

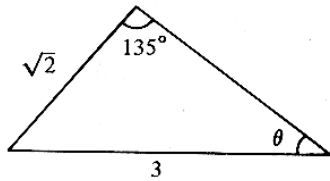
A. $\frac{1}{3}$

B. $\frac{1}{\sqrt{8}}$

D. $\sqrt{\frac{2}{7}}$

C. $\frac{3}{8}$

E. $\frac{1}{\sqrt{2}}$



47 In the figure, if θ is the angle between the diagonals AG and BH of the cuboid, then

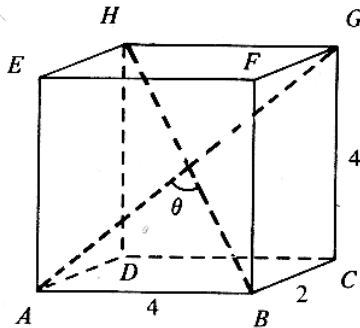
A. $\sin \frac{\theta}{2} = \frac{2}{3}$

B. $\sin \frac{\theta}{2} = \frac{3}{4}$

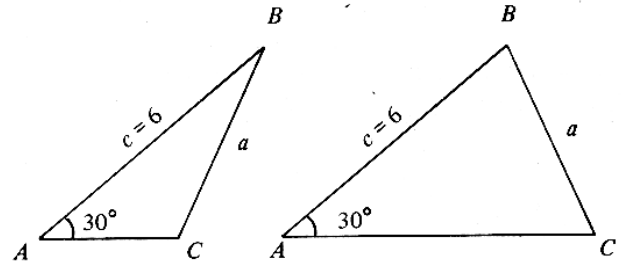
C. $\sin \frac{\theta}{2} = \frac{1}{3}$

D. $\sin \theta = \frac{2}{3}$

E. $\sin \theta = \frac{3}{4}$



49 In $\triangle ABC$, $\angle A = 30^\circ$, $c = 6$. If it is possible to draw two distinct triangles as shown in the figure, find the range of values of a .



A. $0 < a < 3$

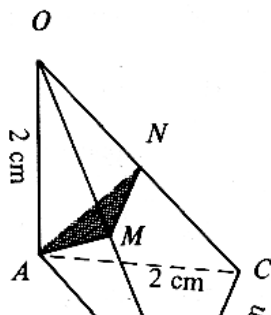
B. $0 < a < 6$

C. $3 < a < 6$

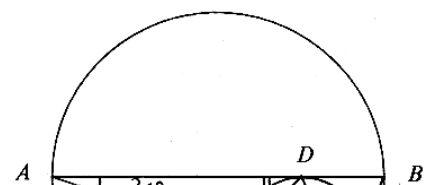
D. $a > 3$

E. $a > 6$

48 In the figure, OA is perpendicular to the plane ABC . $OA = AB = AC = 2\text{cm}$ and $BC = 2\sqrt{2}\text{cm}$. If M and N are the mid-points of OB and OC respectively, find the area of $\triangle AMN$.



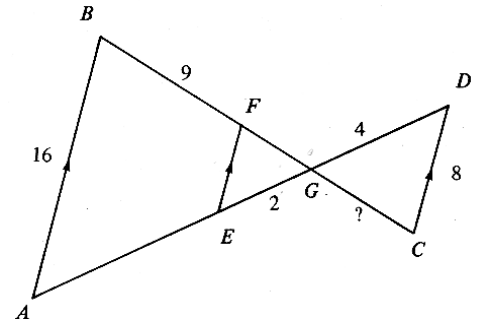
50 In the figure, the two circles touch each other at C . The diameter AB of the bigger circle is tangent to the smaller circle at D . If DE bisects $\angle ADC$, find θ .



- A. 24°
 B. 38°
 C. 45°
 D. 52°
 E. 66°

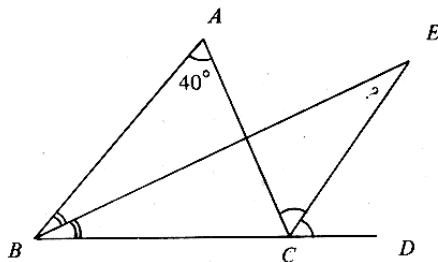
- 53 In the figure, $AB = 16$, $CD = 8$, $BF = 9$, $GD = 4$, $EG = 2$. Find GC .

- A. 4.5
 B. 5
 C. 6
 D. 8
 E. 10



- 51 In the figure, EB and EC are the angle bisectors of $\angle ABC$ and $\angle ACD$ respectively. If $\angle A = 40^\circ$, find $\angle E$.

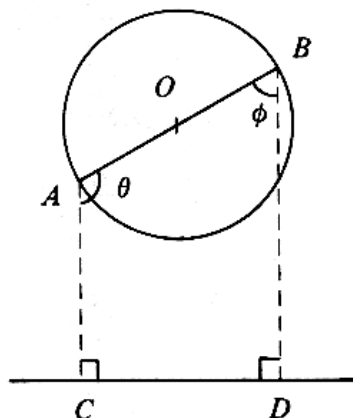
- A. 20°
 B. 25°
 C. 30°
 D. 35°
 E. 40°



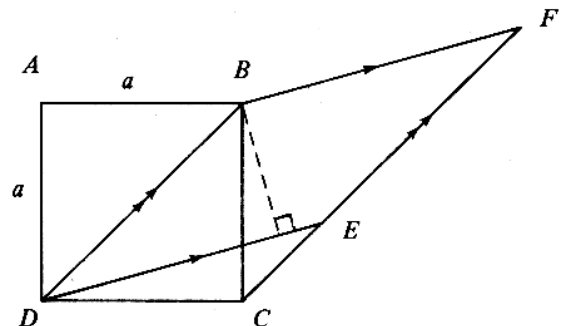
- 52 In the figure, O is the center of the circle. If the diameter AOB rotates about O , which of the following is/are constant?

- I. $\theta + \phi$
 II. $AC + BD$
 III. $AC \times BD$

- A. I only
 B. II only



- 54 In the figure, $ABCD$ is a square of side a and $BDEF$ is a rhombus. CEF is a straight line. Find the length of the perpendicular from B to DE .



- A. $\frac{1}{2}a$
 B. $\frac{2a}{\sqrt{3}}$
 C. $\frac{a}{\sqrt{2}}$
 D. $\frac{\sqrt{3}}{2}a$
 E. a

END OF PAPER